

Stat 517

Homework # 3

1. Let $Y_1 < Y_2 < Y_3 < Y_4$ be the order statistics of a random sample of size 4 from the distribution having pdf $f(x) = e^{-x}, 0 < x < \infty$, zero elsewhere. Find $P(Y_4 \geq 3)$.
2. Let $Y_1 < Y_2 < Y_3 < Y_4$ be the order statistics of a random sample of size $n = 4$ from a distribution $f(x) = 2x, 0 < x < 1$, zero elsewhere.
 - (a) Find the joint cdf of Y_3 and Y_4 .
 - (b) Find the conditional pdf of Y_3 , given $Y_4 = y_4$.
 - (c) Evaluate $\mathbb{E}(Y_3|y_4)$
3. Let X_1, X_2, X_3 be a random sample from a continuous-type distribution having pdf $f(x) = 2x, 0 < x < 1$, zero elsewhere.
 - (a) Compute the probability that the smallest of X_1, X_2, X_3 exceeds the median of the distribution.
 - (b) If $Y_1 < Y_2 < Y_3$ are the order statistics, find the correlation between Y_2 and Y_3 .
4. Let $Y_1 < Y_2 < Y_3 < Y_4 < Y_5$ denote the order statistics of a random sample of size 5 from a distribution having pdf $f(x) = e^{-x}, 0 < x < \infty$, zero elsewhere. Show that $Z_1 = Y_2$ and $Z_2 = Y_4 - Y_2$ are independent.
5. A die was cast $n = 120$ independent times and the following data resulted: If we use a chi-

Sports Up	1	2	3	4	5	6
Frequency	b	20	20	20	20	40 - b

square test, for what values of b would the hypothesis that the die is unbiased be rejected at the 0.025 significance level?

6. Two different teaching procedures were used on two different groups of students. Each group contained 100 students of about the same ability. At the end of the term, an evaluation team assigned a letter grade to each student. The results were tabulated as follows.

Group	Grade				
	A	B	C	D	F
I	15	25	32	17	11
II	9	18	29	28	16

If we consider these data to be independent observations from two respective multinomial distributions with $k = 5$, test at the 5% significance level the hypothesis that the two distributions are the same (and hence the two teaching procedure are equally effective).

7. Prove the converse of the Theorem of Monte Carlo. That is, let X be a random variable with a continuous cdf $F(x)$. Assume that $F(x)$ is strictly increasing on the space of X . Consider the random variable $Z = F(X)$. Show that Z has a uniform distribution on the interval $(0, 1)$.
8. Suppose we are interested in a particular Weibull distribution with pdf

$$f(x) = \begin{cases} \frac{1}{\theta^3} 3x^2 e^{-x^3/\theta^3} & 0 < x < \infty \\ 0 & \text{otherwise.} \end{cases}$$

Determine a method to generate random observations from this Weibull distribution.